BSI Group BSI Group BSI Group

bsi

<<organization>>

Resources for AI Systems

BSIIndia

Inspiring trust for a more resilient world.

©The British Standards Institution 2024

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

Contents

Document Control 3
Version History 3
Reviewers 3
Background 4
Resources Required 4
People Resources 4
Generic Resources 4
System Specific Resources 6
Data Resources 6
Generic Resources 6
System Specific Resources 6
Tooling Resources 7
Generic Resources 7
System Specific Resources 7
System and Computing Resources 7
Generic Resources 7
System Specific Resources 8

BSIIndia

Document ID	Version and Date	Changes	Updated By	Approved By	Data Classification
IAMS/0003	Version 1.0 April 20XX	New documentation	John Doe	Jane Doe	Internal

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

Document Control

This document is the responsibility of the Author. It is subject to formal change control after the initial approved release (e.g. version 1.0). Change requests must be communicated to the Author in writing. Agreed changes are recorded in the Version History of the document.

Version History

Date	Version	Author	Changes
12/02/20XX	0.1	John Doe	New Document – initial draft
01/04/20XX	1.0	John Doe	Issued to Business

Reviewers

Name	Role	Date
Jane Doe	Risk and Compliance Lead	01/04/20XX

BSIIndia

Document ID	Version and Date	Changes	Updated By	Approved By	Data Classification
IAMS/0003	Version 1.0 April 20XX	New documentation	John Doe	Jane Doe	Internal

 ©The British Standards Institution 2024

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

Background

- In order to deliver secure and ethical AI solutions, it is necessary to identify and document relevant resources required at each stage of the AI system life cycle, as well as those required for maintaining the AI Management System (AIMS) and other AI-related activities.
- The required resources documentation within <<organization>> is tiered as follows:
    - Resources required to operate the AIMS, generic resources required / available for the development of AI solutions and guidance on establishing the required resources for each AI system (this document).
    - System-specific resources required for each system (multiple, system specific documents).
- To illustrate the system-specific resource requirements for each system, this document provides examples from the Healthcare AI system. A separately maintained document, <<reference>> Healthcare AI Resource Requirements, sets out in detail the requirements for the Healthcare AI system.

Resources Required

- The resources required for the operation of the AIMS and to support delivery of AI solutions are categorized as follows:
    - People resources
    - Data resources
    - Tooling resources
    - System and computing resources.

BSIIndia

People Resources

Generic Resources

- The key roles required for delivery of the AIMS and AI solutions are as follows:
    - **Chief AI Officer**: Overall accountable for the delivery of AI, provides executive-level endorsement along with the technical vision and AI guardrails.
    - **AIMS Manager**: Located within Risk and Compliance, owns AIMS and conducts day-to-day operations.
    - **Risk and Compliance**: Manages the risk assessment process, internal audit and management reviews.

Document ID	Version and Date	Changes	Updated By	Approved By	Data Classification
IAMS/0003	Version 1.0 April 20XX	New documentation	John Doe	Jane Doe	Internal

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

- **AI Ethics Lead**: Overall responsible for ethical considerations, monitors data usage for biases, ensuring compliance with fairness and transparency guidelines and delivery of system impact assessments.
- **Quality Assurance Lead**: Responsible for data quality audits and monitors compliance with laws and regulations.
- **Data Management Lead**: Overall responsible for all aspects of data management, protection through encryption, data preprocessing, and metrics for monitoring data quality and data analysis.
- **Head of AI Development**: Responsible for secure and ethical development of AI systems and efficient coding to meet climate change objectives.
- **Model Testing Lead**: Overall responsible for testing and monitoring of AI models.
- **AI Product Leads**: Responsible for AI model and system features, model training, product line specific information, design documentation and environmental impact.

- In addition to AI-specific roles, the following are leveraged from the ISO 27001 information security stream:
  - **Cybersecurity Lead**: Additional responsibility for AI incident management.
  - **Lead Counsel**: Responsible for data privacy and regulations specific to AI.
- More information on responsibilities can be found in the IAMS/0002 Roles and Responsibilities document.
- To fulfill the roles set out above, competencies are required in the following key areas:
  - Ethical AI development
  - AI governance
  - Technology research
  - AI prototyping
  - Data security
  - Analytics and modeling
  - Metadata management
  - Testing automation
  - Software QA
  - Incident response
- Details of the competencies for each role are set out in the IAMS/0020 Competency Matrix.

Document ID	Version and Date	Changes	Updated By	Approved By	Data Classification
IAMS/0003	Version 1.0 April 20XX	New documentation	John Doe	Jane Doe	Internal

 ©The British Standards Institution 2024

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

System Specific Resources

- AI Product Leads will define the resources required for day-to-day operation of individual AI solutions.
- Scalable resources will be required for individual AI solutions to support the roles as set out above, such as AI developers and Data Management Team members, to support the additional load for each solution.

Data Resources

Generic Resources

- The data resources required to support the AIMS are minimal:
    - **Data Provenance Database**: Employed as part of the Data Provenance Process (part of the Data Management Process), the Data Provenance Database stores information relating to source/ownership of data items and the manner in which data items have been processed (further details available in the IAMS/0009 Data Provenance Process).
    - **AI Regulatory and Compliance Database**: Records all AI related regulatory and legislative information, managed by the Legal Team, and used by them and other teams, such as Risk and Compliance.

System Specific Resources

- The data resources required for individual systems are defined as part of the development process. These will need to include all significant data sources and data stores employed by the system. For example, for Healthcare AI:
    - Medical database:
        - Medical information database, designed to be the golden source of all medical knowledge, drawn from third party medical archives and databases.
        - Used by Healthcare AI and the Predictive Analysis Engine to formulate diagnoses based on patient data and consultation data (separate databases), obtained from patients by healthcare professionals.
        - Data sources:
            - PubMed
            - Cochrane Library
            - Web of Science
            - Scopus
            - SciFinder Web
            - PsycINFO.

Inspiring trust for a more resilient world.

 ©The British Standards Institution 2024

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

- Data feeds are not real-time, but rather refreshed regularly to ensure the latest medical information is available.
- Research and diagnosis information from Healthcare AI is fed back into the Medical Database to enhance the dataset based on real-world applications continuously. This proprietary information is held separately in the database so that third-party sourced data can be refreshed without impacting the proprietary data.

Tooling Resources

Generic Resources

- The key components of the technology stack and supporting tooling employed by <<organization>> for the development of AI solutions are:
    - **Machine Learning Frameworks**: TensorFlow, PyTorch, and Scikit-learn.
    - **Data Management:** Hadoop and Spark are employed for data processing and management. ProvValidator employed to validate the provenance information for data items and ProvExplorer to visualize the provenance landscape for data items.
    - **Cloud Platforms:** AWS and Azure are employed for scalable computational resources.
    - **Development Environments:** Relying on Jupyter, RStudio, and MATLAB.
- Note that support systems employed by <<organization>> in support of general business operations, such as the WorkDay HR system, do not need to be considered in the context of this document.

System Specific Resources

- The AI Development Team does not use system-specific tooling. Any tooling brought onboard for a new AI solution development will be added to the wider, generic toolset as above.
- AI Product Leads will identify any solution specific to tooling required to support the operation of the system.
- There is currently no solution specific tooling in place.

System and Computing Resources

Generic Resources

- The AIMS document store is held on SharePoint, in Azure.
- The databases employed for the AIMS are held on Azure virtual servers, as follows:
    - Data Provenance Database:

BSI Group BSI Group BSI Group

BSI Group BSI Group BSI Group

        - Database size: 10GB.
        - Low utilization, with expansion rate approx. 10MB pcm.
    - AI Regulatory and Compliance Database:
        - Database size: 1GB.
        - Low utilization, with expansion rate approx. 50KB pcm.
    - Utilization thresholds for these databases has been set to 95% (automatically monitored), at which point database size can be increased.

System-Specific Resources

- AI Product Leads should work with the AI Development Team and IT Team to specify the resource requirements for specific AI solutions.
- AI solutions are operated on cloud platforms (Azure or AWS). The IT Team is able to bring additional virtual servers on stream to meet processing needs as individual solution usage increases.
- Solution specific resource requirements need also to be reflected in the resource requirements for the Predictive Analysis Engine. As more solutions are brought online, or existing solutions expand, the load on the Predictive Analysis Engine needs to be assessed and additional system resources (virtual servers) allocated as necessary.
- Storage requirements for databases supporting each solution need to be documented as part of the solution design documentation. For example, for Healthcare AI:
    - Medical database:
        - Instance: production
        - Version: v8.5, dated 14/08/20XX
        - Size: 10TB
        - Expected expansion rate (approx./pcm) 100MB.

BSIIndia

Document ID	Version and Date	Changes	Updated By	Approved By	Data Classification
IAMS/0003	Version 1.0 April 20XX	New documentation	John Doe	Jane Doe	Internal

BSI Group BSI Group BSI Group